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SHORT COMMUNICATION

The Soil Microbiome of Urban Parks: First Results of Metabarcoding

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ABSTRACT

The soils of urban parks perform important ecosystem functions, regulating runoff, storing carbon and supporting plant growth, yet their microbial communities have been studied far less than those of natural soils. This short communication presents the first results of metabarcoding of bacterial communities in the soils of six parks in an anonymised large city that differ in age and intensity of recreational use. Seventy-two samples of the topsoil were collected from lawns, areas under tree cover and along footpaths. Community composition was determined by sequencing the V4 variable region of the 16S rRNA gene; pH, organic carbon content, soil bulk density and heavy metal concentrations were measured in parallel. Bacterial alpha diversity was highest under tree cover in older parks and lowest along compacted paths. Community structure differed significantly between site types (PERMANOVA, $R^2 = 0.21$) and to a lesser extent between parks. The most important factors shaping community composition were pH and bulk density, whereas the effect of heavy metals at the concentrations studied was weak. These data suggest that park management and layout influence soil microbial diversity at least as much as pollution does. Work is continuing with an expanded sample and the inclusion of fungal communities.

Keywords: urban soils; microbiome; metabarcoding; 16S rRNA; urban parks

Микробиом почв городских парков: первые результаты метабаркодинга

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АННОТАЦИЯ

Почвы городских парков выполняют важные экосистемные функции — регулируют сток, депонируют углерод, поддерживают рост растений, — однако их микробные сообщества изучены значительно хуже, чем сообщества природных почв. В кратком сообщении представлены первые результаты метабаркодинга бактериальных сообществ почв шести парков условного крупного города, различающихся возрастом и интенсивностью рекреационной нагрузки. Отобрано 72 образца верхнего горизонта почвы с газонов, участков под древесной растительностью и вдоль пешеходных дорожек. Состав сообществ определён секвенированием вариабельного участка V4 гена 16S рРНК; параллельно измерены pH, содержание органического углерода, плотность почвы и концентрации тяжёлых металлов. Альфа-разнообразие бактерий оказалось максимальным под древесной растительностью в старых парках и минимальным вдоль уплотнённых дорожек. Структура сообществ достоверно различалась между типами участков (PERMANOVA, $R^2 = 0{,}21$) и в меньшей степени — между парками. Наиболее значимыми факторами, определяющими состав сообществ, были pH и плотность почвы, тогда как влияние тяжёлых металлов в исследованных концентрациях оказалось слабым. Полученные данные свидетельствуют о том, что режим содержания и планировка парков влияют на почвенное микробное разнообразие не меньше, чем загрязнение. Работа продолжается с расширением выборки и включением грибных сообществ.

Ключевые слова: городские почвы; микробиом; метабаркодинг; 16S рРНК; городские парки

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1 INTRODUCTION

Soil microorganisms govern the cycling of carbon and nitrogen, soil structure and plant health. In cities, soils are subject to strong anthropogenic pressure — compaction, contamination, the introduction of fill materials — yet parks, especially old ones, can serve as reservoirs of microbial diversity.

The aim of this communication is to present the first data on the composition and diversity of bacterial communities in urban park soils and to identify the main factors determining their structure.

2 MATERIALS AND METHODS

In six parks (P1–P6) aged from 15 to 120 years, four composite samples were collected from the 0–10 cm layer in each of three site types (lawn, woody vegetation, path verge). DNA was extracted using a commercial kit, and the V4 region of the 16S rRNA gene was amplified and sequenced. The sequences were processed to resolve amplicon sequence variants; diversity was assessed using the Shannon index, and differences in community structure by PERMANOVA on a Bray–Curtis distance matrix.

3 RESULTS

The Shannon index differed between site types (Fig. 1): the highest values were recorded under woody vegetation and the lowest along paths, where soil bulk density reached 1.55 g/cm^3. The results of the PERMANOVA are given in Table 1.

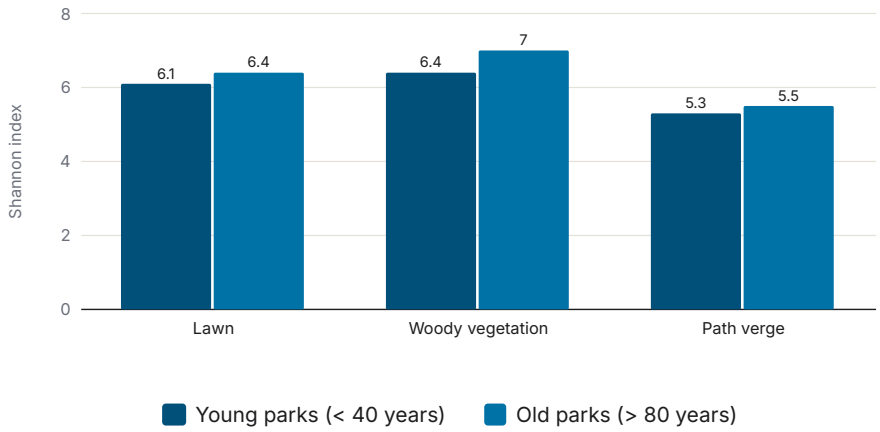


Fig. 1. Shannon index of bacterial communities by site type in young and old parks

Table 1. Factors in the structure of bacterial communities (PERMANOVA, 999 permutations)

Factor	R^2	F	p
Site type	0,21	9,4	0,001
Park	0,12	2,1	0,003
pH	0,09	8,0	0,001
Soil bulk density	0,07	6,3	0,001
Total heavy metals	0,02	1,6	0,092

4 DISCUSSION

The strong influence of site type and soil bulk density points to recreational compaction as the leading factor in the impoverishment of microbial communities. This is consistent with data for natural recreational areas. The weak influence of heavy metals may be explained by their moderate concentrations in the parks studied.

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The practical conclusion is that a well-organised network of paths and the protection of tree root zones from trampling help to preserve soil biodiversity.

5 CONCLUSION

The bacterial diversity of urban park soils is determined primarily by site type, pH and soil compaction. Old parks with preserved tree stands are important reservoirs of soil microbial diversity.

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